

Amplitude-Robust Qubit Spectroscopy Using Echo-Root-Lorentzian Pulse Shapes

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We present a qubit spectroscopy method based on Lorentzian-derived drive pulses that maintains narrow spectral features at drive amplitudes exceeding those used in conventional spectroscopy by more than three orders of magnitude, thereby enhancing robustness against amplitude variations. Specifically, we employ a root-Lorentzian envelope with a π -phase inversion at its peak, which introduces an echo-like symmetry that cancels on-resonant Rabi oscillations while preserving the spectral response. This approach suppresses conventional power broadening and enables high-resolution spectroscopy using pulses shorter than durations required by standard techniques. We experimentally demonstrate this method on a superconducting qubit platform and observe qualitative agreement with numerical simulations, highlighting its potential for rapid and calibration-robust qubit frequency characterization.

Accurate frequency characterization of a quantum two-level system (qubit) is essential for calibrating high-fidelity control in fault-tolerant quantum processors. In many physical implementations, such as superconducting circuits and semiconductor quantum dots, the qubit is realized within an inherently multilevel structure with discrete energy levels, allowing two levels to be selected to form an effective computational subspace that mimics atomic systems via their quantized energy spectrum [1, 2]. Spectroscopic identification of the transition frequency is therefore routinely performed by applying resonant control fields and monitoring the resulting population dynamics [3–5]. Spectroscopic resolution in direct population spectroscopy of coherently driven two-level systems is constrained by decoherence. In the absence of technical noise, stochastic phase fluctuations induce homogeneous broadening of the transition frequency, leading to a minimum resolvable linewidth that scales inversely with the coherence time T_2 , as described by the Bloch equations [6, 7]. This establishes the so-called T_2 limit for conventional direct spectroscopy.

In practice, however, driven spectroscopy introduces an additional trade-off between signal robustness and spectral resolution. When finite-amplitude control pulses are applied, the system dynamics deviate from the idealized two-level response due to coherent Rabi oscillations and coupling to noncomputational levels. Increasing the drive amplitude improves signal contrast and robustness against state-preparation and measurement errors, but simultaneously induces power broadening, resulting in a linewidth that scales with the instantaneous Rabi frequency. Consequently, conventional square-pulse and continuous-wave spectroscopic techniques must operate at low drive amplitudes to maintain resolution approaching the decoherence-limited linewidth set by T_2 , creating a severe practical trade-off between precision and robustness. Smooth temporal envelopes can weaken this broadening, and powers-of-Lorentzian pulses have been shown theoretically and experimentally to exhibit power narrowing [8, 9].

Ramsey interferometry provides a standard alternative to direct spectroscopy and, in principle, enables frequency estimation at the T_2 limit. Achieving such resolution, however, requires long interrogation times and sufficiently dense temporal sampling to resolve the resulting interference fringes. Moreover, Ramsey measurements are susceptible to systematic biases: asymmetries in the exponential decay envelope—arising, for example, from pulse miscalibration—can shift the apparent resonance peak in the frequency domain. Additional effects such as phase fluctuations or unintended AC Stark shifts during the control pulses introduce further distortions, ultimately biasing the extracted transition frequency [10, 11].

Alternative protocols can evade a probe- T_2 resolution bound by using inter-shot phase correlations, external clock references, dynamical decoupling, or tailored filter functions [12–17]. These approaches address different sensing settings and typically require repeated correlations, extended interrogation times, a stable external reference, or additional control resources. Here, we address the resolution-robustness trade-off within direct driven population spectroscopy by employing adiabatically engineered Lorentzian-derived drive pulses with a π -phase inversion at their peak. Unlike prior Lorentzian-pulse power-narrowing demonstrations, this echo-like structure suppresses on-resonant Rabi oscillations while preserving the off-resonant spectral response, maintaining narrow frequency features at drive amplitudes exceeding those used in conventional spectroscopy by more than three orders of magnitude.

Precise determination of the qubit transition frequency is crucial for implementing high-fidelity quantum gates. The drive frequency must coincide with the qubit resonance, as detuning directly reduces gate fidelity. A rigorous metric for quantifying such calibration errors is the diamond norm, which captures deviations in the implemented quantum operation from the ideal one. Minimizing these detuning-induced errors is essential to meet the fault-tolerance thresholds required for scalable quantum computation. [18]

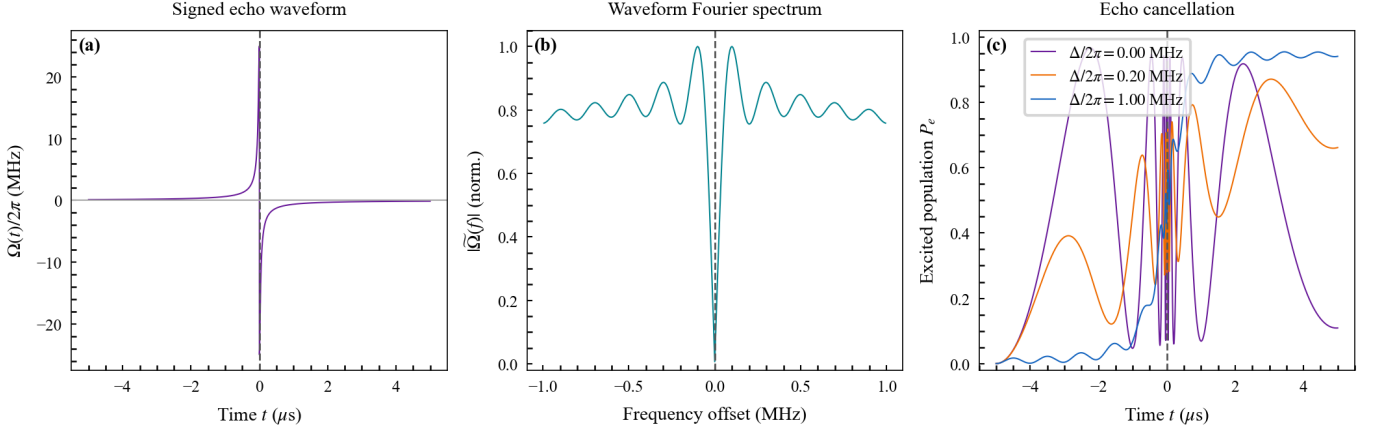


Figure 1. Pulse waveform and echo-cancellation mechanism. (a) Signed echo-root-Lorentzian drive envelope with a π phase inversion at the pulse midpoint. (b) Normalized magnitude of the waveform Fourier spectrum, showing the spectral null at zero frequency produced by the phase inversion. (c) Simulated excited-state population during the pulse at resonance and at non-resonant detunings of $\Delta/2\pi = 0.20$ and 1.00 MHz, using $T_1 = 9999 \mu\text{s}$. At resonance, the evolution accumulated during the first half is reversed during the second half, whereas finite detuning prevents complete cancellation.

Here, we propose a spectroscopic method for probing the qubit transition frequency using arbitrarily shaped drive pulses, specifically an echo-root-Lorentzian pulse. Previous theoretical work showed that Lorentzian pulses exhibit power narrowing, complementary to the conventional power broadening observed for constant drives, as a consequence of their adiabatic properties [8]. Power narrowing with powers-of-Lorentzian pulses was subsequently demonstrated experimentally on IBM Quantum, including the cutoff broadening introduced by truncating the pulse wings [9]. In this work, we extend this framework to longer pulse durations approaching the T_1 timescale and introduce a π -phase jump at the pulse peak. The resulting echo property effectively cancels the net pulse area while preserving the adiabatic properties of the drive. This phase inversion reverses the coherent phase accumulation at resonance, suppressing on-resonant excitation while maintaining an off-resonant spectral response across the frequency domain, and thus making resonance distinguishable.

We define the echo-root-Lorentzian drive from the order- n generalized Lorentzian envelope as

$$L^{(n)}(t) = \frac{1}{\left[1 + \left(\frac{t}{\tau}\right)^2\right]^n}, \quad (1)$$

$$L_{\text{echo}}^{(1/2)}(t) := L^{(1/2)}(t) \text{sgn}(t). \quad (2)$$

where τ sets the characteristic pulse duration and n controls the algebraic decay of the wings. Throughout the theory, Δ and Ω are angular frequencies. Plotted axes show the corresponding cyclic frequencies $\delta = \Delta/(2\pi)$ and $f_R = \Omega/(2\pi)$. We denote an angular-frequency full width at half maximum by Γ_ω and its value in hertz by $\Gamma_f = \Gamma_\omega/(2\pi)$.

For the untruncated, plain generalized Lorentzian

and $n > 1/2$, the usual adiabatic-breakdown estimate gives [8]

$$\Delta_b \tau \propto (\Omega_0 \tau)^{-\frac{1}{2n-1}}, \quad (3)$$

where Δ_b is the angular detuning at which adiabatic following ceases and Ω_0 is the peak angular Rabi frequency. The exponent grows without bound as $n \rightarrow 1/2^+$, so within its domain this asymptotic result predicts the steepest power-narrowing trend at the root-Lorentzian boundary. Exactly at $n = 1/2$, however, Eq. (3) is singular and cannot be evaluated: it does not predict a zero linewidth or an infinite resolution. Finite truncation, decoherence, and the echo phase inversion regularize the response, and the FWHM Γ_f used below is therefore obtained from the complete simulated or measured spectrum. This interpretation is consistent with the power narrowing and cutoff broadening observed for powers of Lorentzian pulses by Mihov and Vitanov [9]. In contrast, a square pulse exhibits conventional power broadening, with $\Gamma_\omega \propto \Omega_0$ at large drive. The derivation and the marginal $n = 1/2$ limit are detailed in the Supplemental Material [19]. Numerical spectroscopy maps for matched root-Lorentzian and echo-root-Lorentzian pulses are shown at broad and narrow detuning scales in Fig. 2. All three rows use $L = 20 \mu\text{s}$; the Lorentzian-derived pulses use $c = 0.002$, and the columns span 50 and 1 MHz, respectively, with matched model parameters. Independent measurements on qubit q1 on 2026-06-28 give $T_1 = 51.24 \pm 0.73 \mu\text{s}$ and a Ramsey time $T_2^* = 7.31 \pm 0.13 \mu\text{s}$ (Supplemental coherence-characterization section). No Hahn-echo decay was acquired for this data set, so the Letter does not assign the Ramsey result to a measured Hahn-echo T_2 . For a conservative measurement-linked transverse reference, we set $T_{2,\text{eff}} = T_2^* = 7.31 \mu\text{s}$, corresponding

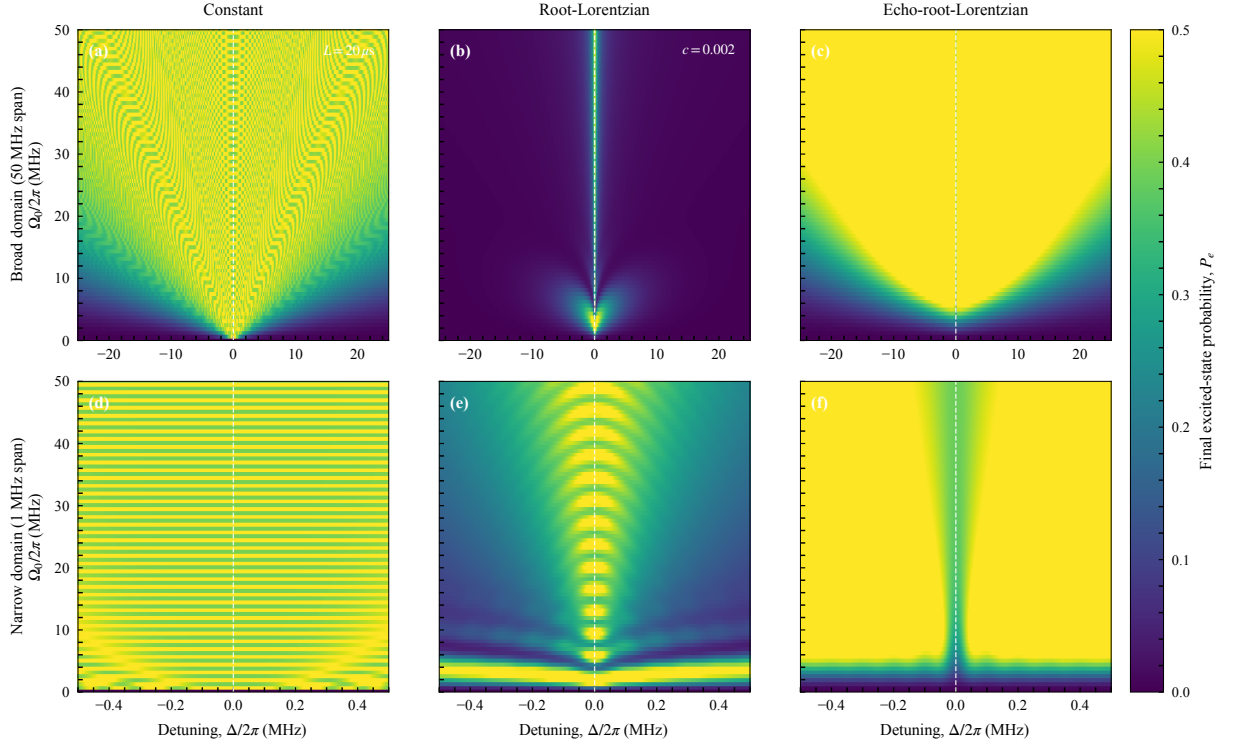


Figure 2. Numerical spectroscopy at two frequency scales for $20 \mu\text{s}$ pulses. The upper row shows the broad 50 MHz domain (-25 to $+25$ MHz), and the lower row shows the narrow 1 MHz domain (-0.5 to $+0.5$ MHz). From left to right, the columns show the constant, root-Lorentzian, and echo-root-Lorentzian protocols; the shaped pulses use relative endpoint amplitude $c = 0.002$. All six panels use the same coherence parameters, peak-Rabi grid, and probability scale. The comparison exposes conventional power broadening for the constant pulse, amplitude-dependent coherent structure for the ordinary root-Lorentzian pulse, and the narrow resonance-centered depletion feature preserved by the echo phase inversion.

to $\Gamma_{f,T_2} = 1/(\pi T_{2,\text{eff}}) = 43.5$ kHz. The calculation in Fig. 2 uses these measurement-linked coherence parameters; $T_{2,\text{eff}}$ is not presented as independent Hahn-echo evidence.

Square pulses exhibit pronounced power broadening even at low drive amplitudes compared to Lorentzian-type pulses, with substantial linewidth expansion observed over three orders of magnitude in amplitude. In contrast, the root-Lorentzian pulse displays the characteristic power-narrowing behavior and maintains a narrow linewidth. However, for pulse durations comparable to the qubit lifetime T_1 , coherent oscillations arise for both square and root-Lorentzian pulses, resulting in amplitude-dependent linewidth variations that hinder precise spectral extraction.

Introducing a π phase inversion at the peak of the root-Lorentzian envelope partitions the drive into two sequential segments of opposite phase, such that the corresponding forward and backward evolutions cancel exactly at zero detuning. This echo-like structure suppresses resonant Rabi oscillations while preserving the off-resonant spectral response, producing a sharp depletion feature centered at the qubit transition frequency. In the frequency domain (Fig. 1), this manifests as a

splitting of the spectral response accompanied by a null at resonance, rendering the measured linewidth largely insensitive to drive amplitude. This mechanism can be viewed as a Loschmidt-echo-like evolution, in which reversibility is achieved only on resonance, while detuning or decoherence breaks the cancellation and restores excitation [20, 21]. Combined with the adiabatic following ensured by the Lorentzian envelope, this interference effect suppresses both power broadening and coherent Rabi oscillations, enabling narrow spectroscopy over a broad range of drive strengths and with pulse durations substantially shorter than those required by standard spectroscopy or Ramsey interferometry. Numerical state-evolution maps illustrating the resonant cancellation and detuned response are presented in the Supplemental Material [19].

Figure 3 resolves the two-dimensional maps into fixed-amplitude spectra. The three rows compare the same peak Rabi frequencies for the ordinary and echo protocols, making the echo-induced central depletion and its persistence with increasing drive amplitude explicit.

In Fig. 4, we plot the reciprocal linewidth as the normalized resolution $R = \Gamma_{f,T_2}/\Gamma_f$. The linewidth of the measured echo response is extracted from Gaussian fits.

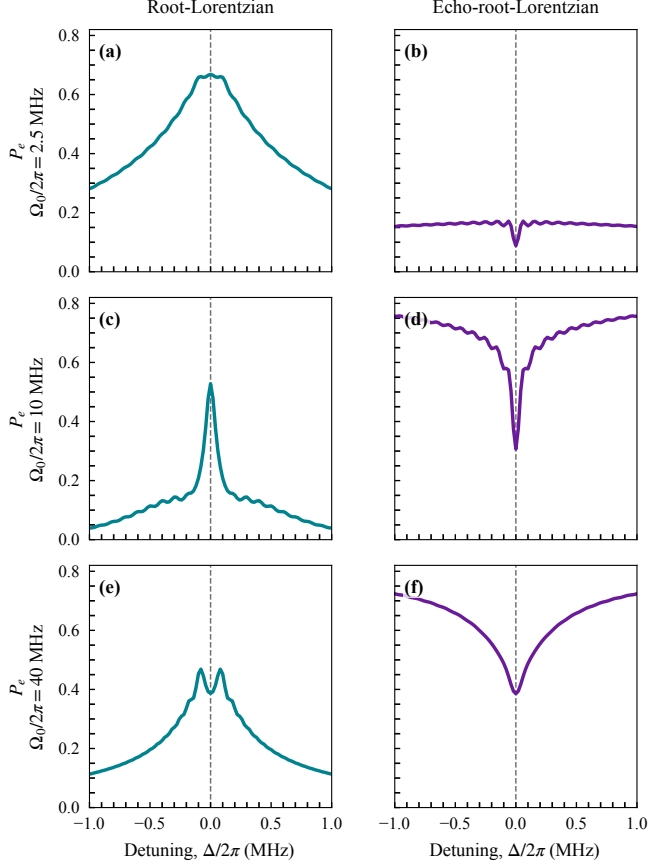


Figure 3. Simulated final excited-state probability P_e versus detuning for 20 μs root-Lorentzian (left) and echo-root-Lorentzian (right) pulses with $c = 0.002$. From top to bottom, the matched peak Rabi frequencies are $\Omega_0/2\pi = 2.5, 10$, and 40 MHz. The ordinary root-Lorentzian spectra develop amplitude-dependent coherent structure, whereas the echo spectra retain a resonance-centered depletion feature over the same drive range. All panels use identical coherence parameters, detuning grids, and vertical scales; dashed lines mark zero detuning.

Although the spectral lineshape of a driven two-level system is often expected to follow a Lorentzian profile, the interference-based structure of the present protocol produces a more complex response. In this regime, Gaussian fitting provides a robust and consistent linewidth estimate within our resolution requirements.

We compare the measured echo-root-Lorentzian metrics from the two-dimensional frequency-amplitude sweep with parameter-matched simulations of both the ordinary and echo root-Lorentzian pulses. The constant-drive reference is the analytic steady-state Torrey solution evaluated using the same measurement-linked T_1 and effective T_2 . Panel (b) reports $Q = AR$, which penalizes a narrow response when its contrast is too small. The three curves therefore expose the useful amplitude window of each protocol directly, including the oscillatory operating structure of the ordinary root-Lorentzian

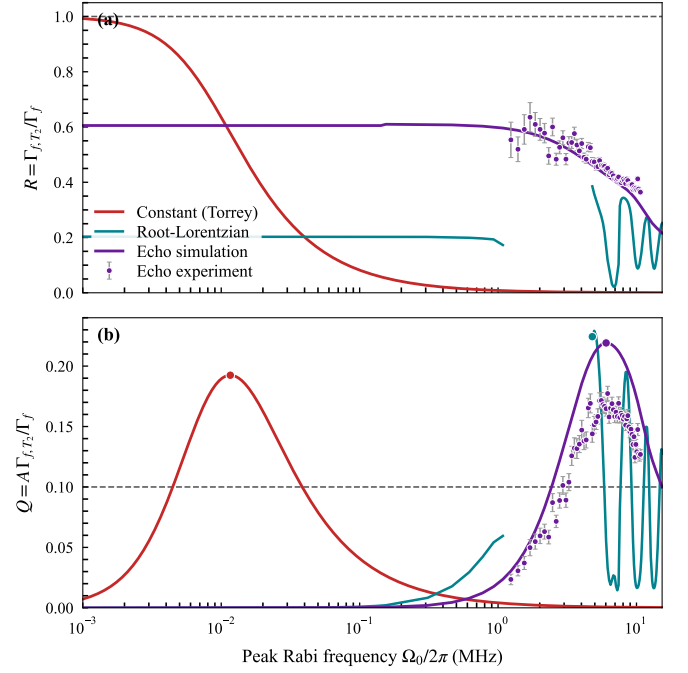


Figure 4. Resolution-contrast comparison for constant, root-Lorentzian, and echo-root-Lorentzian spectroscopy versus peak Rabi frequency Ω_0 . (a) Normalized resolution $R = \Gamma_{f,T_2}/\Gamma_f$, where both quantities are FWHM values in hertz and $\Gamma_{f,T_2} = 1/(\pi T_{2,\text{eff}})$. The red curve is the analytic steady-state Torrey result for a constant drive. The teal curve is the ordinary root-Lorentzian simulation, while purple markers are the measured echo-root-Lorentzian values and the same-colored curve is its parameter-matched two-level Lindblad simulation. Both shaped-pulse models use the recorded 10 μs duration and $c = 0.005$. The dashed line marks the T_2 -limited reference $R = 1$. Additional model-only points extend the simulations below the first nonzero experimental amplitude, and finite model fits continue above the accepted experimental-fit range; the experimental quality gates are unchanged. (b) Contrast-weighted resolution $Q = AR$. Here A is the fitted peak contrast for the ordinary root-Lorentzian pulse, the fitted dip contrast for the echo protocol, and the steady-state population contrast for the constant drive. The dashed line marks the operational threshold $Q = 0.10$; circular curve markers identify each protocol's maximum accepted Q . Experimental error bars are propagated from the local fit covariance, including the contrast-linewidth covariance in panel (b).

pulse and the broader high-amplitude window retained by the echo protocol.

To suppress residual power broadening at large drive amplitudes, the pulse envelope must satisfy requirements on its cutoff behavior. Truncation introduces a residual endpoint drive $\Omega_c = c\Omega_0$ and a corresponding cutoff-broadening contribution [9, 22, 23]. The relevant weak-saturation condition is $s_c = \Omega_c^2 T_1 T_2 \ll 1$, or $\Omega_c \ll 1/\sqrt{T_1 T_2}$; the inequality therefore points toward *smaller*, not larger, endpoint amplitudes. The relative cutoff c cannot be specified independently of Ω_0 , T_1 , and T_2 . Our

cutoff sweeps and high-amplitude measurements are reported in the Supplemental Material [19].

At the core of assessing pulse quality and identifying optimal operating regimes lies the spectral response of the driven qubit. We find that the echo pulse yields spectroscopic features with markedly reduced linewidths over a well-defined region of parameter space. We therefore choose pulse durations shorter than the qubit lifetime T_1 but long enough that Fourier broadening does not dominate.

This constitutes an additional advantage of the echo protocol. While both constant and simple Lorentzian pulses exhibit coherent Rabi oscillations that persist even for pulse durations comparable to or exceeding T_1 , the intrinsic π -phase inversion of the echo pulse suppresses these coherence-induced oscillations. As a result, the spectral response remains smooth and well-behaved, enabling reliable linewidth extraction in regimes where standard and Lorentzian pulses display pronounced oscillatory distortions; see the additional high-amplitude data in the Supplemental Material [19].

Finally, the use of formally infinite-support pulses introduces a three-dimensional design space comprising pulse duration, drive amplitude, and cutoff level. The corresponding cutoff-amplitude simulations and their resolution-signal analysis are reported in the Supplemental Material [19]. In conclusion, we have demonstrated that the echo-Lorentzian pulse constitutes a promising and practical approach for high-resolution spectroscopy. By combining adiabatic shaping with an intrinsic echo phase inversion, the protocol significantly enhances robustness against drive-amplitude variations. Remarkably, it broadens the operational amplitude window by approximately three orders of magnitude, while maintaining high spectral resolution even at drive strengths approaching neighboring qubit transitions. These properties make the echo-Lorentzian protocol particularly attractive for precision qubit characterization under realistic experimental constraints.

This opens the route to coherent spectroscopy based on phase-sensitive off-resonant response rather than population transfer.

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